



全球领先的物联网终端及无线数据方案提供商

MEIG_SLM332A Series Firmware Download Guide

Release date: August 16, 2024

Controlled file name: MEIG-SLM332A Series Firmware Download Guide

Controlled version number: V1.0.0

Publishing organization: Data Division of MeiG Intelligent Technology Co., Ltd.



Important Notice

Copyright Notice

Copyright: Meige Intelligent Technology Co., Ltd.

This document and all its contents are owned by MeiG Intelligent Technology Co., Ltd. and are protected by the laws of China and the copyright laws of applicable international conventions. No one may copy, disseminate, distribute, modify or otherwise use part or all of the contents of this document in any form without the written authorization of MeiG Intelligent Technology Co., Ltd. Violators will be held accountable according to law.

Disclaimer

MeiG Intelligent Technology Co., Ltd. makes no representations or warranties, express or implied, regarding any content in this document, and assumes no liability for merchantability or fitness for a particular purpose or any indirect, special or consequential damages.

Confidentiality Statement

The information contained in this document (including any attachments) is confidential information. The recipient understands that the document is confidential and shall not be used for any purpose other than the specified purpose, nor shall it be disclosed to any third party.

Disclaimer

The company does not assume any liability for property or personal injury caused by abnormal operation of the customer. Please develop corresponding products according to the technical specifications and reference designs in the manual. Before the announcement, the company has the right to change the contents of this manual according to the needs of technological development, and the changed version will not be notified separately.

About the Document

Revision History

	Date	illustrate
Revision V1.0	August 16, 2024	Create initial version

Table of contents

About the Document.....3

1 Download Tool.....5

 1.1 Download Interface.....5

 1.2 Download Process Introduction.....8

Meig Confidential

1Download the tool

about-tools-2024.02.23-win-x64.exe

about-tools-2024.02.23-win-x64.exe > about-tools-2024.02.23-win-x64

名称	修改日期	类型
config	2024/2/23 22:32	文件夹
docs	2024/2/23 22:38	文件夹
drivers	2024/2/23 22:38	文件夹
images	2024/2/23 22:32	文件夹
locales	2024/2/23 22:38	文件夹
platform	2024/2/23 22:38	文件夹
swiftshader	2024/2/23 22:38	文件夹
upload	2024/2/23 22:38	文件夹
about.dll	2024/2/23 22:38	应用程序扩展
about.exe	2024/2/23 22:38	应用程序
about.json.example	2024/2/23 22:38	EXAMPLE 文件
about.log	2024/8/17 15:08	文本文档
adownload.exe	2024/2/23 22:38	应用程序
arelease.exe	2024/2/23 22:38	应用程序

1.1 Download interface

ASR

AboutDownload (Version: 2024.02.23)

DownloadReleaseMisc

Release package:

D:\Users\hanyijia\Desktop\ASR_DDC72E0_20240819_A01_01.zip

Start >

☐ Production mode

☐ Save log

☒ Display log

☒ USB device only

☒ Auto enable for USB

☒ AT Fallback

Action after finished:

Boot continue

Baud rate:

115200

Partition table

Name	Type	Start	Size	vStart	vSize
all	group	0x0000000000	0x0000400000	0xFFFFFFFF	0x0000400000
qspi	flash	0x0000000000	0x0000400000	0x0080000000	0x0000400000
bootloader	ubi	0x0000000000	0x0000016000	0x0080000000	0x0000016000
system	part	0x0000016000	0x000027E000	0x0080016000	0x000027E000
ptable	raw	0x0000016000	0x0000001000	0x0080016000	0x000000088C
fwcerts	raw	0x0000017000	0x0000003000	0x0080017000	0x00000007F4
rd	raw	0x000001A000	0x0000008000	0x008001A000	0x0000000800
apn	raw	0x0000022000	0x0000008000	0x0080022000	0x00000006B84
cp	raw	0x000002A000	0x0000246000	0x008002A000	0x0000246000
dsp	raw	0x0000270000	0x000001D000	0x0080270000	0x000001C6A1
rabin	raw	0x000028D000	0x0000001000	0x008028D000	0x0000000080B
logo	raw	0x000028E000	0x0000006000	0x008028E000	0x000000058AD
reserved	raw	0x0000294000	0x0000102000	0x0080294000	0x0000102000

(1) Function introduction

Release package: Select the version to be burned

Production Mode: Production mode, full erase download is performed by default; it is used during factory production and is not enabled during debugging.

Save Log: Save the tool log of the download process. The log file is saved in the application folder by default.

about-tools-2024.02.23-win-x64.exe > about-tools-2024.02.23-win-x64		
名称	修改日期	类型
swiftshader	2024/2/23 22:38	文件夹
upload	2024/8/19 14:29	文件夹
about.dll	2024/2/23 22:38	应用程序扩展
about.exe	2024/2/23 22:38	应用程序
about.ison.example	2024/2/23 22:38	EXAMPLE
about.log	2024/8/19 14:53	文本文档
adownload.exe	2024/2/23 22:38	应用程序
arelease.exe	2024/2/23 22:38	应用程序
ASR_CRANEL_EVB_CRANEL_A0_04...	2024/8/19 14:31	WinRAR Z
ASR_CRANEL_EVB_CRANEL_A0_04...	2024/8/19 14:27	WinRAR Z
credits.html	2024/2/23 22:38	Chrome H
d3dcompiler_47.dll	2024/2/23 22:38	应用程序扩展
debug.log	2024/7/20 15:49	文本文档
ffmpeg.dll	2024/2/23 22:38	应用程序扩展
icudtl.dat	2024/2/23 22:38	DAT 文件

- Display Log:** Whether to display the tool log during the download process.
- USB device only:** Whether to check only USB devices.
- Auto enable for usb:** For USB devices, automatically determine the download port and enter Download mode after booting.
- AT Fallback:** After the module is powered on, the tool sends AT Command to RESET. If Auto enable for USB is enabled, it can automatically Enter download mode.
- Action after finished:** The action to be performed after the download is completed: (1) No action: No action (2) Reboot: Reboot (3) Boot:
- Turn on the computer.
- Baud rate:** baud rate, default is 115200.

(2) Partition table

Partition table

Name	Type	Start	Size	vStart	vSize
all	group	0x0000000000	0x0000400000	0xFFFFFFFF	0x0000400000
qspi	flash	0x0000000000	0x0000400000	0x0080000000	0x0000400000
bootloader	ubi	0x0000000000	0x0000016000	0x0080000000	0x0000016000
system	part	0x0000016000	0x000027E000	0x0080016000	0x000027E000
ptable	raw	0x0000016000	0x0000001000	0x0080016000	0x000000088C
fwcerts	raw	0x0000017000	0x0000003000	0x0080017000	0x00000007F4
rd	raw	0x000001A000	0x0000008000	0x008001A000	0x00000008000
apn	raw	0x0000022000	0x0000008000	0x0080022000	0x00000006B84
cp	raw	0x000002A000	0x0000246000	0x008002A000	0x0000246000
dsp	raw	0x0000270000	0x000001D000	0x0080270000	0x000001C6A1
rfbin	raw	0x000028D000	0x0000001000	0x008028D000	0x0000000080B
logo	raw	0x000028E000	0x0000006000	0x008028E000	0x000000058AD
reserved	raw	0x0000294000	0x0000102000	0x0080294000	0x0000102000
fota_param	raw	0x0000396000	0x0000003000	0x0080396000	0x00000003000
updater	raw	0x0000399000	0x0000010000	0x0080399000	0x0000000F8E8
fota_pkg	raw	0x00003A9000	0x0000001000	0x00803A9000	0x00000001000
nvm	raw	0x00003AA000	0x0000004E000	0x00803AA000	0x0000004E000
factory	part	0x00003F8000	0x0000008000	0x00803F8000	0x00000008000
factory_a	raw	0x00003F8000	0x0000008000	0x00803F8000	0x00000008000

Download the firmware partition information

Start: partition start address Size: partition size vStart and vSize need to be combined with the Partitions file and are generally ignored.

(3) Download command table

Download command table

Group	Command	Name Partition	Value Image	Weight	Image size
0	require	product	arom-tiny		
0	require	version	0.5		
0	require	version-bootrom	2021.10.16		
0	progress	weight	5857280		
0	call	image	preboot.img		0x6184(24964)
0	call	image	flasher.img		0x7FF0(32752)
0	partition	image	flashinfo.bin		0x1F8(504)
0	partition	image	partition.bin		0x88C(2188)
1	flash	bootloader	bootloader.ubi	176128	0x15000(86016)
2	flash	ptable	partition.bin	8192	0x88C(2188)
2	flash	fwcerts	fwcerts.bin	16384	0x7F4(2036)
2	flash	rd	ReliableData.bin	65536	0x8000(32768)
2	flash	apn	apn.bin	61440	0x6B84(27524)
2	flash	cp	cp.bin	4767744	0x246000(2383872)
2	flash	dsp	dsp.bin	237568	0x1C6A1(116385)
2	flash	rfbin	rf.bin	8192	0x80B(2059)
2	flash	logo	logo.bin	49152	0x58AD(22701)
2	flash	updater	updater.bin	131072	0xF8E8(63720)
3	erase	partition	fota_param	12288	
3	erase	partition	fota_pkg	4096	
4	erase	partition	nvm	319488	
5	erase	partition	factory	32768	

In the Download command table, we can configure the operations of each partition file. You can configure in the blue box on the left to choose whether to burn.

It is worth noting that nvm and factory

☒	4	erase	partition	nvm	319488
☐	5	erase	partition	factory	32768

nvm will delete calibration parameters by default during download after checking

After factory is checked, the initialization configuration will be deleted by default during download; it is only checked by default in Production Mode, and cannot be selected in non-Production Mode

1.2 Download process introduction

1. Open the upgrade tool:

Enter the about-tools-2024.02.23-win-x64 tool package and open "about.exe

2. Select the corresponding firmware compressed file

Release package:

3. Configure the corresponding download options

☐ Production mode

☐ Save log

☒ Display log

☒ USB device only

☒ Auto enable for USB

☐ AT Fallback

Action after finished:

No action

Baud rate:

115200

This configuration will automatically enter the download mode after booting and complete the download process

Note: Checking the nvm option will erase the calibration parameters, and it is not necessary to check it during debugging.

☒	3	erase	partition	tota_param	12288
☒	3	erase	partition	fota_pkg	4096
☒	4	erase	partition	nvm	319488
☐	5	erase	partition	factory	32768

4. Download process

ASR AboutDownload (Version: 2024.02.23)

Download Release Misc

D:\Users\hanyijia\Desktop\ASR_DDC72E0_20240819_A01_01.zip

Save Training Order Back

1 | COM4

1 COM4 0% RUNNING

```
17:29:48.107 extracting download.json (2 KB)...
17:29:48.107 download engine running in upgrade mode!
17:29:48.107 about download engine started successfully.
17:29:48.107 getting serial devices list...
17:29:51.202 <COM4> new device arrived.
17:29:51.202 enabling device <COM4> into downloading mode...
17:29:51.202 device <COM4> enabled successfully.
17:29:51.202 connecting to serial device <COM4>...
17:29:51.234 <COM4> connected to serial device <COM4> successfully!
17:29:51.234 <COM4> starting to fire device <COM4>...
17:29:51.235 <COM4> ONLINE device_open_failed_num = 0, device_list size = 1
```

5. Download completed

ASR AboutDownload (Version: 2024.02.23)

Download Release Misc

D:\Users\hanyijia\Desktop\ASR_DDC72E0_20240819_A01_01.zip

Save Training Order Back

1 | COM4

1 COM4 SUCCEEDED

```
17:30:03.747 <COM4> (flasher) program "logo" done.
17:30:03.747 <COM4> (flasher) programming partition 'updater' with datap = #0x7e200000, size = 0x10044
17:30:04.194 <COM4> (flasher) program "updater" done.
17:30:04.194 <COM4> (flasher) erasing partition 'fota_param'
17:30:04.199 <COM4> (flasher) erase "fota_param" done.
17:30:04.199 <COM4> (flasher) erasing partition 'fota_pkg'
17:30:04.200 <COM4> (flasher) erase "fota_pkg" done.
17:30:04.200 <COM4> (flasher) erasing partition 'nvm'
17:30:04.539 <COM4> (flasher) erase "nvm" done.
17:30:04.539 <COM4> OKAY [ 0.860s]
17:30:04.539 <COM4> all finished. total time: 12.155s
```